A Study of Post-Quantum Digital Signatures in the Automotive Industry

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***Abstract*— Digital signature algorithms widely used in the automotive industry, such as RSA, ECDSA, and EdDSA, face imminent security risks from quantum computing. To address this, NIST initiated the standardization of post-quantum cryptographic algorithms. CRYSTALS-Dilithium, a leading candidate, was standardized in August 2024 as FIPS 204 and NIST designated it as the Module-Lattice-based Digital Signature Algorithm (ML-DSA). This work evaluates the applicability of post-quantum digital signature algorithms (PQ-DSAs) for automotive systems. We benchmark CRYSTALS-Dilithium on two automotive-grade evaluation boards: the SIMCom SIM8918EA development platform with a Qualcomm ARM Cortex-A53 processor, and the Infineon AURIX TriBoard TC-222. Based on performance measurements, we assess the feasibility of potential integration of Dilithium into automotive use cases.**

Keywords—post-quantum, cryptography, digital signatures, ECU, automotive, CRYSTALS-Dilithium, ML-DSA

I. INTRODUCTION

The automotive industry is seeing a shift towards software-defined vehicles (SDV). This shift is accompanied by an increase in digital communications which need to be secured and authenticated. This authentication is currently provided by "pre-quantum" or classical digital signature schemes such as RSA, ECDSA and EdDSA. The long lifecycles of road vehicles warrant that these secure communications remain secure for long periods of time. However, Shor proposed an algorithm to significantly reduce the time needed to break public-key and symmetric-key cryptographic algorithms respectively [1].

The field of quantum computing is seeing rapid development. Quantum computers are predicted to solve certain complex problems exponentially faster than classical computers. Tasks that would take today's supercomputers thousands of years could potentially be completed in hours or even minutes on a large-scale, error-corrected quantum computer. For example, the RSA digital signature scheme is based on the integer factorization problem while ECDSA and EdDSA are based on the mathematical hardness of the discrete logarithm problem, which a quantum computer running Shor's algorithm can solve very efficiently. Quantum computers capable of fully utilizing Shor's algorithm to break public-key cryptographic algorithms and, by extension, perform signature forgery/tampering attacks are predicted to become available in the next 10 years. This could enable the injection of malicious firmware or software into vehicle ECUs, among other threats, thereby compromising the safety of the vehicle and its passengers. This is where post-quantum cryptography comes into play: there is a need to be prepared for the future. The US White House issued a memorandum setting 2035 as the primary target for transitioning cryptographic systems to quantum-resistant algorithms. The quantum risk can be assessed using three parameters: the security shelf life of information assets (typically very large in case of automotives), the time required to migrate to post-quantum cryptosystems, and the time remaining before quantum computers break current cryptosystems. The transition period from classical cryptosystems to post-quantum cryptosystems needs to be minimized to mitigate the risk of classical cryptosystems being exposed to post-quantum attack scenarios.

PQC algorithms have been implemented, benchmarked and optimized across diverse hardware platforms in the context of various domains. Some of the hardware platforms worked on previously include field-programmable gate arrays (FPGAs) [2], graphics processing units (GPUs) [3], central processing units (CPUs) [4], and resource-constrained microcontrollers. Such evaluations are essential for understanding the practical feasibility, performance trade-offs, and hardware suitability of post-quantum schemes in real-world deployment scenarios in different domains. However, there are still research gaps in the implementation and performance measurements on automotive-grade hardware.

CRYSTALS-Dilithium, hereafter referred to simply as "Dilithium", is part of the CRYSTALS suite of cryptographic algorithms and is one of the post-quantum digital signature algorithms (PQ-DSA) recently standardised by the National Institute of Standards and Technology (NIST) as FIPS 204 – Module Lattice-Based Digital Signature Algorithm (ML-DSA) [5]. It is one of the top choices and is predicted to be secure against attacks by quantum computers of the future.

This paper aims to contribute insights into the feasibility and challenges associated with migration to post-quantum cryptosystems in the context of the automotive industry. Practical integration strategies, roadmaps and benchmarking of algorithms other than Dilithium are out of the scope of this paper.

- The digital signature algorithms currently used in cars today are briefly discussed.
- The use cases of asymmetric digital signature algorithms in vehicular networks are enumerated.

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- Upcoming PQ-DSA standards which show promise for use in the automotive context are analyzed and briefly compared.
- In this work, we measure key performance metrics of the Dilithium algorithm implemented on a SIMCom SIM8918EA evaluation board (used for development of infotainment systems among other use cases) as well as an Infineon AURIX TC-222 automotive-grade microcontroller (used for Classic AUTOSAR applications) and discuss the feasibility of integrating Dilithium into automotive use cases.

II. Post-Quantum Digital Signatures in Automotives

A. Application of Digital Signatures in Vehicles

In the automotive world, digital signature algorithms are currently used in a variety of use cases such as secure FOTA (Firmware Over-The-Air) updates, where firmware for various ECUs (Electronic Control Units) of the car are transmitted over the air to the car's gateway ECU for remote flashing. In this scenario, forged signatures pose obvious security risks such as an attacker potentially being able to remotely flash unauthenticated malicious firmware to an ECU of the car over the air, thus gaining access to the car's critical systems or causing damage to the car and/or injury or loss of life to the driver and passengers of the car. Other areas where digital signatures are used within road vehicles include secure boot, V2X communication, secure diagnostics, vehicle identification and secure vehicle access. A survey on PQC migration in automotive details current use cases and standards relying on classical cryptography, identifying suitable candidates for PQC adoption. These include automotive standards such as ISO 15118 for plug and charge, AUTOSAR for software architecture, and IEEE 1609.2 for vehicle-to-vehicle (V2V) communication [6].

B. Brief About Dilithium

Dilithium is widely considered to be the frontrunner among PQ-DSAs available today when considering the balance of security levels, key and signature footprints, resource requirements and versatility. It is a module lattice-based algorithm which employs the LWE (Learning with Errors) problem and is resilient, in theory, to attacks by quantum computers [4]. NIST chose Dilithium in the fourth round of its PQC standardization process as the preferred digital signature algorithm due to its balance between speed, key size, and security [5]. Dilithium and FALCON were also the only PQC signature schemes found to be suitable for integration into the TLS 1.3 [7].

Dilithium has three parameter sets, which are Dilithium2, Dilithium3, Dilithium5. In general, the different variants provide different key sizes and security levels, with Dilithium 5 having a NIST security level of 5 being the most secure variant. The trade-off is reduced speed of signature generation and verification and larger key and signature sizes when compared to the smaller instances Dilithium3 and Dilithium2. The sizes of the public keys, private keys and signatures for each of the three variants of Dilithium are depicted in Table I.

The Dilithium algorithm includes three main processes, namely key generation, signature generation and signature verification. The key generation, signing and verification algorithms for Dilithium are explained in Fig. 1.

Key generation: The keypair generation algorithm takes two random seeds ρ and K as inputs and generates the matrix A and two vectors s_1 and s_2. From these it derives t and finally generates the keypair consisting of a secret key, sk and a public key, pk as the final output, which are used for signing and verifying processes.

Signing algorithm: The signature generation algorithm takes the message M and the secret key sk as inputs to generate the signature. During the signing process it compute w= Ay, where y is the masking vector. The high order bits w_1 is then hashed with message to generate the challenge c. This challenge goes as input to create the signature z, where z = cs_1+ y. Subsequently, the signing algorithm calculates the hints h which will be used for verifying the signature. Finally, the Sign (sk, M) algorithm sends $\sigma = (z, h, c)$ as the signature to the receiver's side.

Verifying algorithm: The signature verification algorithm generates $w_1^{'}$ which is then hashed with the message M. The hashed value obtained is the challenge c. If the calculated c matches with the one received in signature and the norm z is also valid, then signature is valid

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1) $\rho \leftarrow \{0,1\}^{256}$
2) $K \leftarrow \{0,1\}^{256}$
3) $(s1, s2) \leftarrow S_n^l \times S_n^k$
4) $A \in R_q^{k \times l} :=$ ExpandA(ρ)
5) $t := As_1 + s_2$
6) $(t_1, t_0) := Power2Rouind_q(t, d)$
7) $tr \in \{0,1\}^{384} :=$ CRH(ρ II t_1)
8) $return\ (pk = (\rho, t_1),$ sk = $(\rho, K, tr, s_1, s_2, t_{01}))$

Sign(sk, M)

9) **A** $\in R_q^{k \times l} :=$ ExpandA(ρ)
10) $\mu \in \{0,1\}^{384} :=$ CRH(tr II M)
11) K := 0, (z, h) := $\perp$
12) *While* (z, h) = $\perp$ *do*
13) y $\in S_{r1-1}^l :=$ ExpandMask(K II μ II k)
14) **w** := **Ay**
15) $w_1 := HighBits_q(w, 2\gamma_2)$
16) $c \in B_{60} := H(\mu$ II $w_1)$
17) z := y $+cs_1$
18) $(r_1, r_0) := Decompose_q(w - cs_2, 2r_2)$
19) If ||z|| $||z||_\infty \geq \gamma_1 - \beta\ or\ ||r_0||_\infty \geq \gamma_2 - \beta\ or\ r_1 \neq w_1$, then (z,h) := $\perp$
20) Else
21) h := MakeHint$_q(-ct_0, w - cs_2 + ct_0, 2\gamma_2)$
22) if $||ct_0||_\infty \geq \gamma_2$ or the #of 1's in h is greater than ω, then (z,h) := $\perp$
23) $K := K + 1$
24) **return** $\boldsymbol{\sigma = (z, h, c)}$

Verify (pk, M, $\boldsymbol{\sigma = (z, h, c)}$**)**

25) $A \in R_q^{k \times l} :=$ ExpandA (ρ)

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26) $\mu \in \{0,1\}^{384} \coloneqq \mathrm{CRH}(\mathrm{CRH}(\rho||t_1)||M)$
27) $w_1' \coloneqq \mathrm{UseHint}_q(h, Az - ct_1 2^d, 2\gamma_2)$
28) return $[\![||z||_\infty < \gamma_1 - \beta]\!]$ and $[\![c = \mathrm{H}(\mu||w_1')]\!]$ and $[\![\#\text{of 1's in h is} \leq \omega]\!]$

Fig. 1. Dilithium signature scheme [4]

TABLE I. DILITHIUM KEY AND SIGNATURE SIZES

Variant	Metric		
	Public Key Size	***Private Key Size***	***Signature Size***
Dilithium2	1312 bytes	2560 bytes	2420 bytes
Dilithium3	1952 bytes	4032 bytes	3309 bytes
Dilithium5	2592 bytes	4896 bytes	4627 bytes

[a.] Dilithium key and signature sizes for all three variants

C. Other Post-Quantum Digital Signature Schemes

Aside from Dilithium, two other post-quantum digital signatures were also selected for standardization by NIST as part of the standardization process. These are SPHINCS+ and FALCON.

- SPHINCS+ is a stateless hash-based digital signature algorithm, standardized by NIST as FIPS 205, Stateless Hash-Based Digital Signature Standard (SLH-DSA) in August 2024 [8]. It is the only hash-based PQ-DSA standard selected to be standardized, with the other top PQ-DSAs, namely Dilithium and FALCON, being lattice-based. Unlike earlier hash-based schemes such as XMSS, SPHINCS+ is stateless, eliminating the need to track signing states such as previously used key pairs—between signatures. SPHINCS+ possesses very small public and private keys as low as 32 bytes, at the cost of very large signature sizes, making it slower and impractical for resource-constrained environments and applications having strong latency requirements.
- FALCON (Fast-Fourier Lattice-based Compact Signatures over NTRU) is another lattice-based PQ-DSA selected by NIST in the fourth round of the post-quantum standardization process. At the time of writing this paper, FALCON is still yet to be standardized, but it is planned to be standardized as FIPS 206 – FN-DSA, short for FFT over NTRU-Lattice-Based Digital Signature Algorithm. FALCON is a great PQ-DSA option for resource-constrained environments like embedded systems, given its outstanding signature verification speed [9] and small public key and signature sizes [10]. However, FALCON signature generation comes with high computational complexity, requiring significant resources compared to Dilithium [7].

III. METHODOLOGY

A. Hardware

For the purposes of benchmarking the algorithm, two automotive-grade evaluation boards serving different purposes within the context of a vehicle were used. The first hardware target for evaluation of the Dilithium algorithm was the Automotive Realtime Integrated NeXt Generation Architecture microcontroller (AURIX) platform. Specifically, an Infineon AURIX TriBoard TC-222 evaluation board was used which boasts 96 kB of internal SRAM, consisting of 88 kB of Data Scratch-Pad RAM (DSPR) and 8 kB of Instruction Scratch-Pad RAM (PSPR). The total Flash memory available on the TC-222 evaluation board is the combination of the Data Flash Memory of 96 kB and Program Flash Memory of 1 MB. It operates at a clock frequency of 133MHz. The board was operated in a single-core bare-metal environment. The hardware setup is depicted in Fig. 2. The second board was the SIMCom SIM8918EA evaluation board, running Android on a Qualcomm chipset with an ARM Cortex A53 processor. The hardware setup for the SIM8918EA board is depicted in Fig 3. This board is used for prototyping vehicle infotainment and OTA applications among other use cases. The clock speed of this board is up to 2 GHz, a significant jump from the TC222 board.

B. Software

The official reference implementation of Dilithium written in C was used for benchmarking purposes on the TC222 board. The TASKING C/C++ toolchain was used with AURIX Development Studio IDE to compile, link, flash, debug and run the program on the TC-222 evaluation board.

An optimized implementation of Dilithium utilizing ARM NEON SIMD instructions was made available by the authors of [11], which was earlier tested on an Apple M1 and Raspberry Pi. However, this implementation has not been tested on an automotive-grade evaluation board to our knowledge. We benchmarked this optimized implementation on the SIM8918EA board and benchmarked the reference implementation as a baseline.

C. Implementation

The Dilithium2 variant, which satisfies NIST security level 2, was selected for performing benchmarks on the TC-222 development board, due to decreased cryptographic overhead, faster processing times and lower memory consumption compared to Dilithium3 and Dilithium5 variants. The larger parameter sets of Dilithium3 and Dilithium5 have higher resource requirements that the TC-222 development board cannot support due to stack size and RAM

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limitations and thus were not benchmarked on the TC222 board.

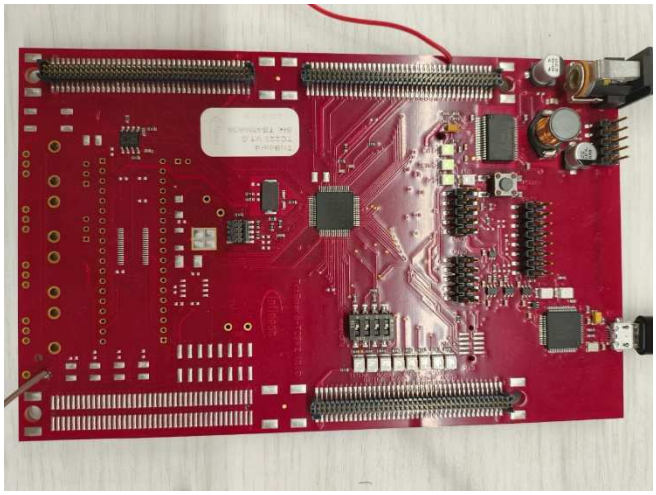


Fig. 2. Infineon AURIX TriBoard TC222 evaluation board

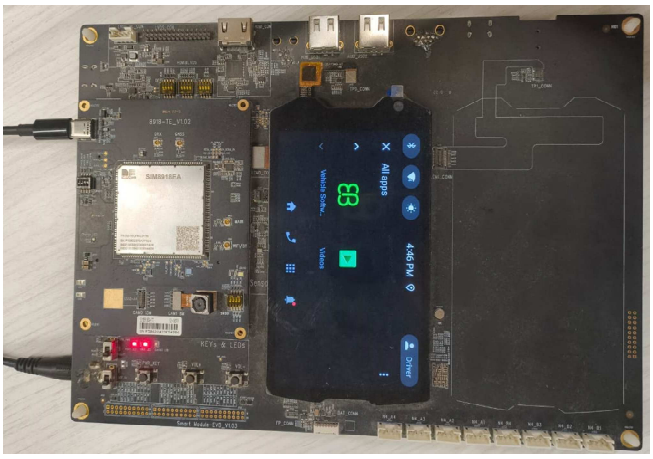


Fig. 3. SIMCom 8918EA evaluation board

The Infineon AURIX TriBoard TC-222 evaluation board was connected to a Windows PC running the AURIX Development Studio IDE via a micro-USB-B cable. Through Infineon's Device Access Server (DAS) middleware, the board can be connected to AURIX Development Studio for building and flashing code, as well as for program execution and debugging using the TASKING C/C++ toolchain.

Tests were conducted using randomly generated messages of varying sizes to simulate diverse use cases and communication protocols typically encountered in vehicular networks. Message authentication on CAN traditionally uses short symmetric message authentication codes (MACs) for low latency and bandwidth. However, certain use cases such as OTA updates, multi-ECU verification and secure diagnostics leverage asymmetric digital signatures despite their higher performance cost.

The CAN protocol specifies the size of a data frame to be 8 bytes, while CAN-FD supports up to 64-byte frames. SOME/IP payloads are virtually unlimited in size, restricted by practical hardware limitations and transport layer payload limitations of UDP and TCP. These message sizes were selected for signing and verification operations using Dilithium2 on the TC-222.

On the SIM8918EA evaluation board, the optimized implementation of Dilithium using ARM's NEON SIMD instructions was benchmarked and its performance was compared to the reference implementations for Dilithium2, Dilithium3 and Dilithium5. The tests were run 30 times for statistical validity and averages of the key generation, signature generation and signature verification times were recorded.

The metrics that were measured as part of the benchmark on the TC-222 and SIM8918EA evaluation boards can be found in Tables II-IV.

Benchmarking of signature generation and verification was also performed with a large message size of 1 GB and hardcoded public and secret keys on the TC222 board, to imitate an OTA firmware update or a large SOME/IP payload. This was made possible despite memory limitations of the TC-222 by streaming the message in small blocks to be absorbed by the SHAKE256 hashing function. The TC-222 MCU is not best suited for such use cases due to its low processing power, but this experiment revealed the possibility of streaming and signing large messages on resource-constrained hardware, and also highlighted some challenges of migration from classical to post-quantum cryptosystems, such as the increased key and signature sizes and also the high computation complexity and resource requirements to sign and verify messages using post-quantum digital signature schemes.

IV. Results

The key performance metrics on the TC222 board, including keypair generation time, signature generation time and signature verification time are shown in Tables II and III.

Table II shows the performance of the Dilithium2 official reference implementation on the TC222 board for varying message sizes to simulate messages of different protocols. It is observed that message size barely affects the time taken for each of the keygen, signature generation and signature verification operations.

The time taken to generate and verify signatures for a 1GB message signed with Dilithium2 on the TC222 board are shown in Table III. The results show the impracticability of using asymmetric post-quantum digital signature schemes for real-time applications on resource-constrained automotive embedded environments.

Table IV shows the benchmark results of the three variants of Dilithium, comparing the baseline reference implementation's performance with the speedup introduced by the ARM-optimized implementation. We observe an average speed up of 1.83x for key generation, 3.65x for signature generation and 2.11x for signature verification, which is highly promising for further platform-specific optimizations.

Dilithium shows promise for automotive use cases as a well-rounded, versatile and secure post-quantum digital signature scheme, and although algorithms such as FALCON have advantages over Dilithium such as faster signature verification speed which is very valuable in automotive embedded systems, Dilithium nevertheless secures its place as a strong contender for use in vehicles.

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TABLE II. DILITHIUM2 METRICS – SMALL MESSAGES

Message length	Key generation time (ms)	Signature generation time (ms)	Signature verification time (ms)
59 bytes (Default)	20.876	112.788	47.187
8 bytes (CAN)	20.837	90.127	46.969
64 bytes (CAN-FD)	20.825	84.755	47.190

[a] Context length of 14 bytes in all benchmarks, which is the default value

TABLE III. DILITHIUM2 METRICS – 1GB MESSAGE

Metric Name	Metric Value
Signature generation time	30.216 minutes
Signature verification time	29.983 minutes
SRAM utilization (KB)	27.8 KB
SRAM utilization (%)	28.96%
Program memory utilization (KB)	64.8 KB
Program memory utilization (%)	6.33%

[a] Metrics measured on TC222 evaluation board

TABLE IV. PERFORMANCE COMPARISON OF DILITHIUM VARIANTS

Scheme	Time Taken (ms)		
	KeyGen	*Sign*	*Verify*
Dilithium2 (ref)	0.450	2.354	1.008
Dilithium2 (NEON)	0.250	0.624	0.480
Dilithium3 (ref)	0.750	3.624	1.591
Dilithium3 (NEON)	0.406	1.030	0.754
Dilithium5 (ref)	1.244	4.487	2.583
Dilithium5 (NEON)	0.68	1.464	1.268

[a] Average metrics across 30 test samples measured on SIMCom SIM8918EA evaluation board with 59 byte randomly generated message

V. CONCLUSION AND FUTURE WORK

Post-quantum algorithms are the next logical step, not only in the world of automotive security, but in the broader domain of information security. Adoption of post-quantum digital signatures is a must for future-proofing vehicles and is the need of the hour. In this paper, we identified some key performance indicators of post-quantum digital signatures and measured them by benchmarking the Dilithium algorithm on automotive development platforms simulating real-life ECUs. We benchmarked Dilithium on the SIMCom SIM8918EA evaluation board running Android on an ARM Cortex A53 processor and achieved significant speedups with an implementation of Dilithium optimized for ARM targets using NEON SIMD instructions. We also benchmarked the lighter Dilithium2 variant on an Infineon AURIX TriBoard TC-222 automotive-grade microcontroller evaluation board. Further, we discussed various use cases for post-quantum digital signature schemes in the automotive domain. It was observed that Dilithium signature generation is more time-consuming than signature verification due to the additional calculation of hints during signature generation. Within the limits of the available memory on the hardware, signature generation and verification speeds remain similar for different message lengths. For a large message size, streaming was used to allow the SHAKE256 algorithm to absorb the message in small blocks before calculating the hash value for signature generation. Dilithium2 seems to be the most suitable for resource-constrained automotive environments given its smaller parameter set. For infotainment ECUs running on ARM-based processors, NEON SIMD instructions provide significant speedups in the keypair generation, signature generation and signature verifications operations of Dilithium. We conclude Dilithium to be a strong contender to consider when migrating automotive cryptosystems to post-quantum cryptography in the near future.

Key challenges in the migration to post-quantum cryptosystems in automotives include:

- Large public and secret key sizes compared to classical digital signature schemes
- High computational complexity
- Backward compatibility with classical cryptosystems
- Vulnerability to side-channel attacks and fault injection attacks (dependant on implementation), as discussed in [12].

Future directions in this field include:

- Migration and integration strategies to ease the adoption of new post-quantum cryptography standards like CRYSTALS Dilithium and other PQ-DSA algorithms into existing protocols and frameworks, such as AUTOSAR SecOC and V2X communication protocols such as Dedicated Short-Range Communication (DSRC).
- Performance optimizations to reduce processing overhead and latency in signature generation and verification.
- To ease the migration to post-quantum cryptosystems, hybrid cryptographic approaches as discussed in [13] can be adopted to provide backward compatibility with classical cryptographic schemes while also extending capabilities of the cryptosystem with post-quantum cryptography algorithms.
- The Dilithium algorithm can be subjected to rigorous stress testing and adversarial analysis to uncover potential vulnerabilities and further strengthen its security guarantees, for example, through customized implementations as discussed in [12].

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